Jiafan He, Ph.D.

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Education

Sep 2019 – Ongoing **University of California, Los Angeles**
Ph.D. in Computer science (Ph.D. Candidate in Mar 2022)
Adviser: Prof. Quanquan Gu

Aug 2015 – Jul 2019 **Tsinghua University**
B.E in Computer science and Technology.

Research interest

- **Machine learning** (Reinforcement learning, Large Language Models), **Optimization**

Award and honors

2015 **Gold medal in 56th International Mathematical Olympiad** (top $\sim$ 6 in china to form a group that represent China and take part in the competition, top $\sim$ 30 in all national team get Gold medal and final rank 10)

2023 **Amazon Science Hub Fellowship** Fall 2023 and Winter 2024 quarters (full tuition and a 10,950/quarter stipend)

2024 **Dissertation Year Award** Summer 2024, Fall 2025 and Sprint 2025 quarters (full tuition and a 6,666/quarter stipend)

Publications

Conference publications

- **Achieving a fairer future by changing the past**
  Jiafan He, Christos-Alexandros Psomas, Ariel D. Procaccia, David Zeng
  International Joint Conference on Artificial Intelligence (IJCAI), 2019
- **Provably efficient reinforcement learning for discounted MDPs with feature mapping**
  Dongruo Zhou, Jiafan He, Quanquan Gu.
  International Conference on Machine Learning (ICML), 2021
- **Logarithmic Regret for Reinforcement Learning with Linear Function Approximation**
  Jiafan He, Dongruo Zhou, Quanquan Gu.
  International Conference on Machine Learning (ICML), 2021
- **Minimax Optimal Reinforcement Learning for Discounted MDPs**
  Jiafan He, Dongruo Zhou, Quanquan Gu.
  Neural Information Processing Systems (NeurIPS), 2021

Publications (continued)

- **Uniform-PAC Bounds for Reinforcement Learning with Linear Function Approximation**
  Jiafan He, Dongruo Zhou, Quanquan Gu.
  Neural Information Processing Systems (NeurIPS), 2021
- **Nearly Optimal Regret for Learning Adversarial MDPs with Linear Function Approximation**
  Jiafan He, Dongruo Zhou, Quanquan Gu.
  International Conference on Artificial Intelligence and Statistics (AISTATS), 2022
- **Learning Stochastic Shortest Path with Linear Function Approximation**
  Yifei Min, Jiafan He, Tiaohao Wang, Quanquan Gu.
  International Conference on Machine Learning (ICML), Baltimore, 2022
- **On the Sample Complexity of Learning Infinite-horizon Discounted Linear Kernel MDPs**
  Yuanzhou Chen, Jiafan He, Quanquan Gu.
  International Conference on Machine Learning (ICML), Baltimore, 2022
- **A Simple and Provably Efficient Algorithm for Asynchronous Federated Contextual Linear Bandits**
  Jiafan He, Tianhao Wang, Yifei Min, Quanquan Gu.
  Neural Information Processing Systems (NeurIPS), New Orleans, 2022
- **Nearly Optimal Algorithms for Linear Contextual Bandits with Adversarial Corruptions**
  Jiafan He, Dongruo Zhou, Tong Zhang, Quanquan Gu.
  Neural Information Processing Systems (NeurIPS), New Orleans, 2022
- **Locally Differentially Private Reinforcement Learning for Linear Mixture MDPs**
  Chonghua Liao, Jiafan He, Quanquan Gu.
  Asian Conference on Machine Learning (ACML), 2022
- **Nearly Minimax Optimal Regret for Learning Linear Mixture Stochastic Shortest Path**
  Qiwei Di, Jiafan He, Dongruo Zhou, Quanquan Gu.
  International Conference on Machine Learning (ICML), Hawaii, 2023
- **Optimal Online Generalized Linear Regression with Stochastic Noise and Its Application to Heteroscedastic Bandits**
  Heyang Zhao, Dongruo Zhou, Jiafan He, Quanquan Gu.
  International Conference on Machine Learning (ICML), Hawaii, 2023
- **On the Interplay Between Misspecification and Sub-optimality Gap in Linear Contextual Bandits**
  Weitong Zhang, Jiafan He, Zhiyuan Fan, Quanquan Gu.
  International Conference on Machine Learning (ICML), Hawaii, 2023

Publications (continued)

- **Nearly Minimax Optimal Reinforcement Learning for Linear Markov Decision Processes**
  Jiafan He, Heyang Zhao, Dongruo Zhou, Quanquan Gu.
  International Conference on Machine Learning (ICML), Hawaii, 2023

- **Cooperative Multi-Agent Reinforcement Learning: Asynchronous Communication and Linear Function Approximation**
  Yifei Min, Jiafan He, Tianhao Wang, Quanquan Gu.
  International Conference on Machine Learning (ICML), Hawaii, 2023

- **Uniform-PAC Guarantees for Model-Based RL with Bounded Eluder Dimension**
  Yue Wu, Jiafan He, Quanquan Gu.
  Uncertainty in Artificial Intelligence (UAI), Pittsburgh, 2023

- **Provably Efficient Representation Selection in Low-rank Markov Decision Processes: From Online to Offline RL**
  Weitong Zhang, Jiafan He, Dongruo Zhou, Amy Zhang, Quanquan Gu.
  Uncertainty in Artificial Intelligence (UAI), Pittsburgh, 2023

- **Variance-Dependent Regret Bounds for Linear Bandits and Reinforcement Learning: Adaptivity and Computational Efficiency**
  Heyang Zhao, Jiafan He, Dongruo Zhou, Tong Zhang, Quanquan Gu.
  Conference on Learning Theory (COLT), 2023

- **Horizon-free Reinforcement Learning in Adversarial Linear Mixture MDPs**
  Kaixuan Ji, Qingyue Zhao, Jiafan He, Weitong Zhang, Quanquan Gu.
  International Conference on Learning Representations (ICLR), 2024

- **Pessimistic nonlinear least-squares value iteration for offline reinforcement learning**
  Qiwei Di, Heyang Zhao, Jiafan He, Quanquan Gu
  International Conference on Learning Representations (ICLR), 2024

- **Towards robust model-based reinforcement learning against adversarial corruption**
  Chenlu Ye, Jiafan He, Quanquan Gu, Tong Zhang
  International Conference on Machine Learning (ICML), 2024

- **A nearly optimal and low-switching algorithm for reinforcement learning with general function approximation**
  Heyang Zhao, Jiafan He, Quanquan Gu
  Neural Information Processing Systems (NeurIPS), 2024

Publications (continued)

- **Achieving Constant Regret in Linear Markov Decision Processes**
  Weitong Zhang, Zhiyuan Fan, Jiafan He, Quanquan Gu
  Neural Information Processing Systems (NeurIPS), 2024

Preprints

- **Accelerated Preference Optimization for Large Language Model Alignment**
  Jiafan He, Huizhuo Yuan, Quanquan Gu
  Neural Information Processing Systems (NeurIPS), Adaptive Foundation Models Workshop 2024

- **Nearly optimal algorithms for contextual dueling bandits from adversarial feedback**
  Qiwei Di, Jiafan He, Quanquan Gu

- **Reinforcement learning from human feedback with active queries**
  Kaixuan Ji, Jiafan He, Quanquan Gu

Professional services

Conference Reviewer for

2021	**AAAI Conference on Artificial Intelligence**
2020,2021,2022,2023,2024,2025	**International Conference on Learning Representations (ICLR)**
2021,2022,2023,2024	**International Conference on Machine Learning (ICML)**
2021,2022,2023,2024,2025	**International Conference on Artificial Intelligence and Statistics (AISTATS)**
2020,2021,2022	**International Joint Conferences on Artificial Intelligence (IJCAI)**
2020,2021,2022,2023,2024	**Neural Information Processing Systems (NeurIPS)**

Visiting graduate student at

2020	**Simons Institute for the Theory of Computing**

Skills

Coding	C, C++, Python